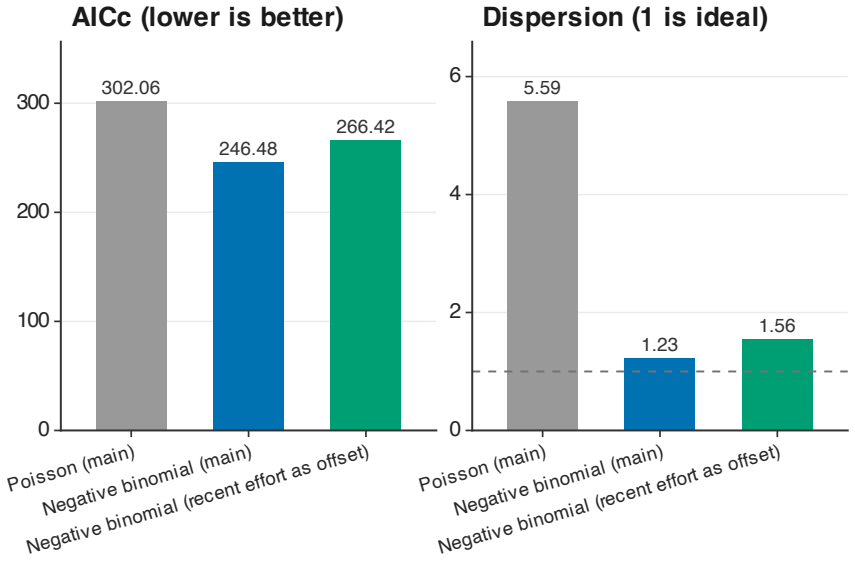
a

Poisson is rejected, negative binomial is not

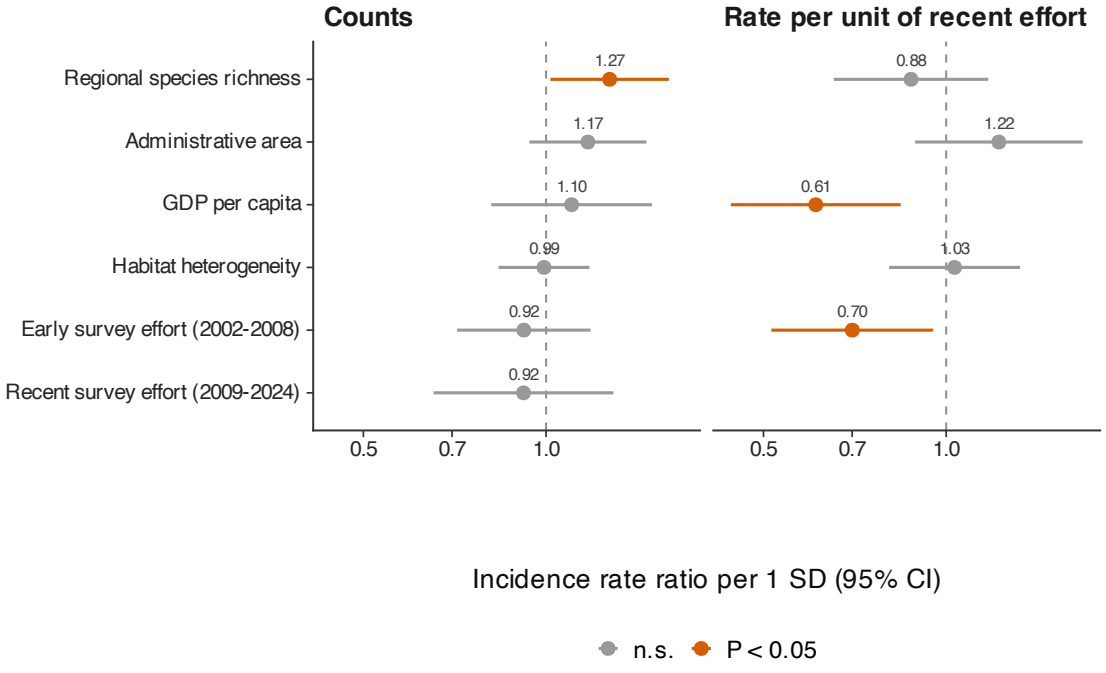
Poisson dispersion 5.60 (P = 2.3e-17); the negative binomial reduces it to 1.24 (P = 0.20)



b

Effort shows up as diminishing returns

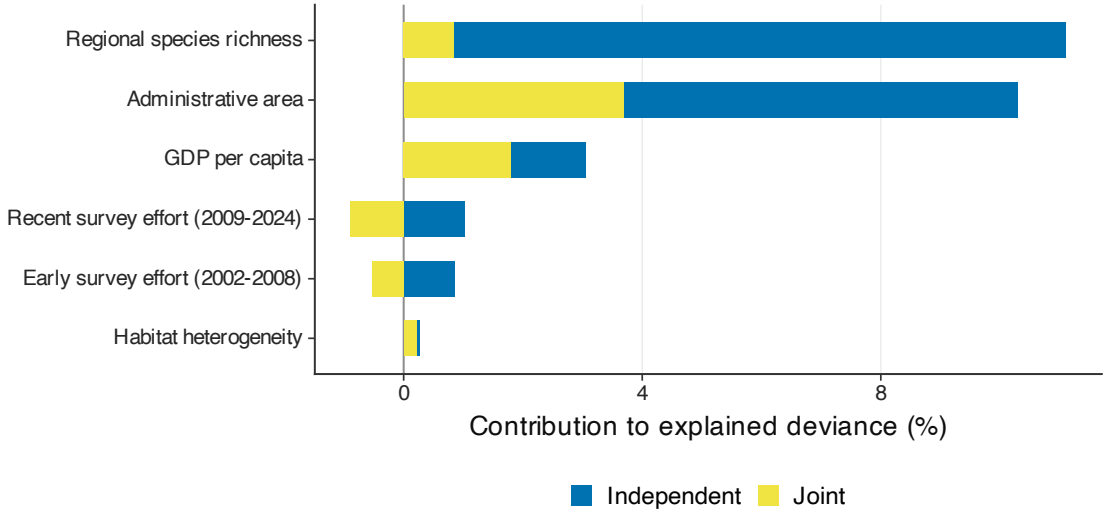
Left: both efforts as predictors. Right: recent effort as an offset, so the response is the discovery rate per unit of effort. Provinces already well covered in 2002-2008 then yield fewer records



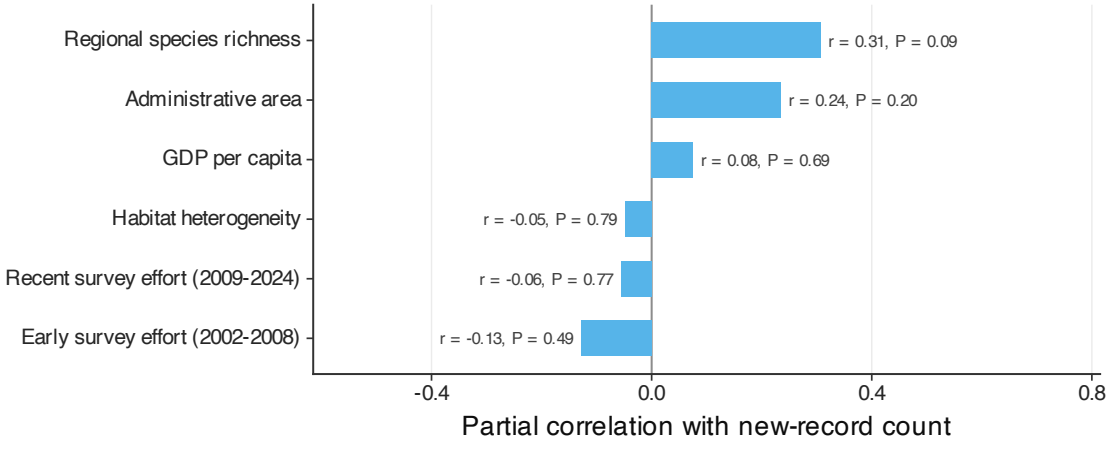
c

Hierarchical partitioning

Full model explains 20.5% of deviance; independent contributions sum to 19.9%



d

Partial regression

Each predictor against the residuals of a model holding all the others constant